

PATRONS

- **Prof. Prasad Krishna**, Director
- **Dr. M Radhakrishnan**, Registrar
- **Prof. P Mohanan**, PIC Academics

PROGRAM CONVENER

- **Dr. Jayakrushna Sahoo**, HoD, Dept. of CSE 1, IIIT Kottayam
- **Dr. Rubell Marion Lincy G**, HoD, Dept. of CSE 2, IIIT Kottayam

CO-ORDINATORS

- **Dr. Vengadeswaran S**
Assistant Professor, Dept. of CSE, IIIT Kottayam
- **Dr. Cinu C Kiliroor**
Assistant Professor, Dept. of CSE, IIIT Kottayam
- **Dr. John Paul Martin**
Assistant Professor, Dept. of CSE, IIIT Kottayam
- **Dr. Priyadharshini S**
Assistant Professor, Dept. of CSE, IIIT Kottayam



ABOUT US

Indian Institute of Information Technology Kottayam (IIIT Kottayam) was established in the year 2015 as one among the IIITs established under the administrative control of the Government of India and later declared as an "Institution of National Importance" by an Act of Parliament enacted in 2017. IIIT Kottayam imparts top-notch technological education at undergraduate, postgraduate and doctoral levels. Undergraduate and postgraduate programmes in Computer Science and Engineering, Electronics and Communication Engineering, AI & DS and Cyber Security, are available at the institute.



**ALL INDIA COUNCIL FOR TECHNICAL
EDUCATION [AICTE]**

**SPONSORED 5 DAY OFFLINE FACULTY
DEVELOPMENT PROGRAMME FOR
FACULTY MEMBERS OF BBA/BCA/BMS
INSTITUTES ON**

**"PRACTICAL DATA SCIENCE AND
ADVANCED MACHINE LEARNING
EXPLORATION
WITH HANDS-ON USING PYTHON:
FROM BASICS TO INSIGHTS"**

[09th - 13th DECEMBER 2024]

Mode: OFFLINE

Free Registration [Sponsored by AICTE]

ORGANISED BY

Dept. of CSE

**Indian Institute of Information
Technology Kottayam, Kerala**



**Indian Institute of
Information Technology
Kottayam**

OBJECTIVES

- To equip participants with a comprehensive understanding with hands-on experience in the concepts of data science, ML and DL

Topics (not limited to)

- Python Basics for Data Science
- Practical Data Science: [Data Types, Merging, Concatenation, Cross-validation, cleaning] Pre-processing - Finding Missing Data, Encoding, Transformation, and Outlier, Feature scaling, Data Wrangling, Data Normalisation
- Machine Learning: Supervised Learning: Classification - Simple and Multiple Linear Regression - Unsupervised Learning : K-Means Clustering, Hierarchical - Reinforcement Learning: Exploring Q-Learning: Applications And Benefits
- Deep Learning: Building and training a Neural Network -Convolutional Neural Networks [CNN]

Highlights

- Hands-on training using Python
- Solving Real-world problem of ML & DL
- Industrial Applications & Case Studies

WHO CAN ATTEND

- Faculty members of BBA/BCA/BMS institutes

PAYMENT

- Registration & Programme FEE: **FREE OF COST**
- Refreshments, Lunch & Dinner will be provided.
- Shared accommodation at the institute will be provided for **FREE OF COST**

• **ONLY LIMITED REGISTRATION AS APPROVED BY AICTE**
• **BASED ON a first-come BASIS**

Registration

REGISTRATION LINK:

<https://forms.gle/eweKyZjNMwHKMp4V8>



Attendance & Assessment

- Attendance: 80% (min.)
- Assessment: Two [MCQ]

SPEAKERS

- Experts from Academia & Industry having strong Background in Practical DS and Advanced ML

CONTACT

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**FDP RELATED QUERIES
EMAIL US AT:**

✉ dsaml.fdp@gmail.com

**LAST DATE FOR REGISTRATION:
02.12.2024**



IIIT KOTTAYAM



Schedule

[09 - 13 DECEMBER 2024]

Mode: OFFLINE

VENUE: IIT KOTTAYAM

Day	Time	Topics
Day 1 09.12.2024 [Monday]	08.30 AM - 09.30 AM	Registration
	09.30 AM - 10.45 AM	Inauguration
	11.00 AM - 12.30 PM	Python Basics for Data Science [Hands-on using Python]
	12.30 PM- 01.30 PM	Lunch Break
	01.30 PM - 03.00 PM	Practical Data Science – Know your data [Data Types, Merging, Concatenation, training and test split, Cross-validation, cleaning]
	03.15 PM - 04.45 PM	Pre-processing - Finding Missing Data, Encoding, Categorical Data, Data Transformation, Outlier. [Hands-on using Python]
Day 2 10.12.2024 [Tuesday]	09.30 AM - 10.45 AM	Feature scaling, Feature Scaling, Data Wrangling, Data Normalization
	11.00 AM - 12.30 PM	Introduction to Machine Learning
	12.30 PM- 01.30 PM	Lunch Break
	01.30 PM - 03.00 PM	Supervised Learning [Classification Algorithm]
	03.15 PM - 04.45 PM	Solving a real-world problem using Classification Algorithm [Hands-on using Python]

Day	Time	Topics
Day 3 11.12.2024 [Wednesday]	09.30 AM - 10.45 AM	Supervised Learning [Linear Regression: Simple and Multiple Linear Regression]
	11.00 AM - 12.30 PM	Solving a real-world problem using Linear Regression [Hands-on using Python]
	12.30 PM-1.30 PM	Lunch Break
	01.30 PM - 03.00 PM	Unsupervised Learning - K-Means Clustering, Hierarchical [Hands-on using Python]
	03.15 PM - 04.45 PM	Assessment - 1 [Quiz]
Day 4 12.12.2024 [Thursday]	09.30 AM - 10.45 AM	Reinforcement Learning
	11.00 AM - 12.30 PM	Exploring Q-Learning: Applications and Benefits [Hands-on using Python]
	12.30 PM-1.30 PM	Lunch Break
	01.30 PM - 03.00 PM	Building and training a Neural Network
	03.15 PM - 04.45 PM	Introduction to Deep Learning
Day 5 13.12.2024 [Friday]	09.30 AM - 10.45 AM	Convolutional Neural Networks [CNN]
	11.00 AM - 12.30 PM	Real-world applications of CNN [Hands-on using Python]
	12.30 PM-1.30 PM	Lunch Break
	01.30 PM - 03.00 PM	Assessment - 2 [Quiz]
	03.15 PM - 04.45 PM	Valedictory